Charliemagne Malicay

Full-Stack Engineer · DevOps & AWS · AI & Automation Engineer

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PROFILE

Full-stack engineer with 11 years across embedded, desktop, R&D and production web platforms, now focused on applied AI. I build in Next.js, React and TypeScript on the front end and Python (FastAPI, Flask, Django) or NestJS on the back end, and I run DevOps end to end: AWS infrastructure as code with Terraform, containerised delivery (Docker, ECS/Fargate), CI/CD pipelines, monitoring and incident response.

My AI work centres on retrieval-augmented generation (RAG) and agent automation: OCR and embedding pipelines into Pinecone and PostgreSQL, LangChain/LangGraph agents, and workflow orchestration in n8n and Make. I care about the parts that keep these systems running: rate-limit fallbacks, output validation and retries, monitoring, and root-cause analysis. AWS Certified Cloud Practitioner; working toward AWS Certified Machine Learning Engineer – Associate.

SELECTED IMPACT

- **RAG assistant used by 20–100 client staff:** led a small team building the Technical Manual AI Bot, cutting the time to find answers in technical manuals from hours to minutes.
- **Production outage diagnosed and fixed:** traced an EC2 out-of-memory outage to OCR ingestion sharing an instance with the API, using Grafana memory spikes and OOM-killer logs. Moved ingestion to a queue-fed worker with batching and verified the fix by re-running a large ingestion.
- **CI/CD and infrastructure as code:** built GitHub Actions pipelines (multi-stage Docker builds, layer caching, OIDC authentication to AWS, build, push to ECR, deploy to ECS) and provisioned AWS with Terraform across Entvas, Icebear and AK Holdings.
- **Database architecture:** designed read/write routing on RDS PostgreSQL (primary plus read replicas, custom SQLAlchemy session routing) so freshness-critical endpoints always hit the primary.
- **LLM resilience:** built automatic fallback to a secondary model on 429s and provider errors, plus validated-and-retried structured output for classification workflows.

EXPERIENCE

Senior Automation Engineer — Ideahub Solution LLC (contract) Jul 2026 – Sep 2026

- Built n8n and Make workflows integrating GoHighLevel APIs and custom GHL-based CRMs, including a lead-intake scenario (landing-page webhook into GHL, error-handler retries) running under 1,000 operations a month.
- Added LLM lead classification (OpenAI, Anthropic): JSON output requested in the prompt, parsed and validated, retried on parse failure, with 429 rate limits handled.
- Orchestrated the Viktor AI Employee and Grok Bot across Slack, Apollo, Clay, Smart Lead and GoHighLevel to run multi-step outbound and CRM processes.
- Built web and mobile apps with Next.js, FastAPI and React Native/Expo on PostgreSQL, Supabase and SQLite; maintained WordPress landing pages.

Full-Stack & AWS Developer — Entvas Apr 2025 – May 2026

- Built Next.js, React, Tailwind and shadcn/ui interfaces using spec-first development (Kiro Spec).
- Designed modular NestJS services with custom interceptors, JWT and role guards, @CurrentUser/@Roles decorators and dynamic modules; tuned MySQL schemas and queries for high-throughput workloads.
- Provisioned AWS with Terraform: ECS, ECR, Fargate, EC2 and Lambda, with Amplify and Cognito for hosting and identity; event-driven pipelines on EventBridge, SQS, SNS and SES; NGINX routing for Next.js and Python deployments.
- Built CI/CD in GitHub Actions: multi-stage Docker builds, layer caching, OIDC authentication to AWS, and build, push to ECR and deploy to ECS pipelines.
- Optimised AWS cost models alongside capacity.

Full-Stack Developer — Icebear Solutions (contract) Feb 2025 – Mar 2025

- Delivered the Football Gallery platform end to end: FastAPI and MySQL back end, Next.js and shadcn/ui front end covering auth, portals, profiles and payments.
- Designed a normalised ERD (users, players, managers, agents, fans); built JWT-authenticated endpoints with filtering, sorting and pagination.
- Integrated Gemini for video analysis, Stripe and PayPal for payments, CSV import pipelines, and a RAG player and team search (Pinecone plus OpenAI).
- Shipped to Ubuntu EC2 behind NGINX with Docker Compose, Terraform, CI/CD, unit and integration tests, Swagger/ReDoc docs, SSL/TLS and automated MySQL backups.

Full-Stack Developer / AI Engineer — AK Holdings (contract) Nov 2024 – Mar 2025

- Led a small team (front end, data prep, QA) on the Technical Manual AI Bot: a RAG chatbot on Next.js and LangServe (FastAPI) with LangChain agents, OpenAI embeddings and Pinecone, used by 20–100 client staff.
- Wrote Python jobs to OCR PDFs and images from S3 into Pinecone, and loaded a 50 GB CSV dataset into PostgreSQL.
- Integrated GPT-3.5, GPT-4 and Claude with automatic fallback to a secondary model on rate limits and errors.
- Designed RDS primary and read-replica routing behind an ALB with custom SQLAlchemy session routing.
- Deployed on EC2 with NGINX and Docker Compose via Terraform; enforced OAuth, JWT, CORS and SSL/TLS; added Pytest and Jest suites, and Prometheus and Grafana monitoring that exposed the production OOM failure described above.

Senior Frontend Developer — Shakewell Agency (contract) May 2024 – Sep 2024

- Built responsive, mobile-first web apps and PWAs in TypeScript and React; improved load times by refactoring components and reducing network requests.
- Worked with designers, PMs and back-end engineers on feature scope; championed accessibility and cross-browser compatibility.

Software Engineer, Level 2 — Innodata Knowledge Inc. Apr 2022 – Mar 2024

- Full-stack engineer in R&D, primarily Python: Flask and FastAPI services and React/Next.js front ends over SQL and NoSQL stores.
- Delivered machine-learning and image-processing solutions, including object-detection models running on in-house GPU machines, and a RAG document-QA pipeline integrating the OpenAI API.
- Wrote design and analysis documentation for stakeholders, provisioned infrastructure with Terraform and automated NGINX-based deployments.

Earlier experience

- **Associate R&D Engineer**, Advanced World Solution Inc. (Oct 2021 – Mar 2022): debugged production defects, managed release deployments, ran simulator and hardware integration tests, documented root-cause analysis in Agile Scrum.
- **Full-Stack Developer**, SCL (Jul 2018 – Jun 2021): Flask REST APIs, Python Lambda functions, React/TypeScript/Electron apps, jQuery-to-React migrations, MySQL and SQLite schemas, API Gateway, SQS and Amplify.
- **Software Developer**, EasyRent (Nov 2017 – Feb 2018): Flask and PostgreSQL REST APIs and a jQuery front end.
- **Embedded Developer**, Ship Electronics (Jun 2017 – Oct 2018): programmed microcontrollers for sensor I/O; designed schematics and PCBs.
- **Software Engineer**, HGG Profiling Specialists Philippines Inc. (Apr 2014 – Jun 2015): Python and Django back-end APIs, jQuery and Knockout.js front ends, automated PDF reporting.

TECHNICAL SKILLS

Languages: Python (11y) · JavaScript (11y) · TypeScript (9y) · HTML5 & CSS3 (11y)

Front end: React (9y) · Next.js (7y) · React Native & Expo (3y) · Tailwind CSS (5y) · Chakra UI (5y) · Material UI (5y) · shadcn/ui (3y) · WordPress (3y)

Back end: Flask (7y) · FastAPI (4y) · NestJS (1–2y) · ExpressJS (3y) · REST APIs (11y) · GraphQL (5y) · Django and Django REST

Databases: PostgreSQL (7y; schema design, Alembic/Django migrations, EXPLAIN-driven tuning, pgvector, RDS operations, 100 GB+) · MySQL (5+y) · SQLite (5y) · MongoDB (4y) · DynamoDB (3+y) · Elasticsearch/OpenSearch (3+y) · Supabase (3y) · Pinecone (2y)

AWS: EC2 (8y) · S3 (8y) · Lambda (8y) · RDS (8y) · Amplify (7y) · ECS/ECR/Fargate (2y) · EventBridge, SQS, SNS, SES (2y) · Cognito (2y) · Azure (basics)

DevOps & CI/CD: Git (11y) · Ubuntu on EC2 (8y) · Docker & Compose (3–4y) · GitHub Actions (3–4y) · Terraform (4y) · NGINX (5y) · tmux (5y) · GitLab CI · Prometheus & Grafana monitoring · automated database backups · SSL/TLS · incident root-cause analysis

Generative AI: OpenAI/GPT (4y) · RAG (4y) · Hugging Face (4y) · LangChain (3y) · embeddings & vector search (3y) · Gemini (3y) · Anthropic Claude (2y) · LangGraph (2y) · LangServe (2y)

Automation & integrations: PDF parsing & OCR (8y) · Slack (4y) · n8n (1y) · Make (1y) · GoHighLevel (1y) · Apollo (1y) · Clay (1y) · Smart Lead (1y)

Security & quality: JWT (8y) · OAuth (7y) · Swagger/ReDoc (9y) · SSL/TLS · CORS · Pytest · Jest

Tooling: Claude Code, Codex, Kiro, Cursor, GitHub Copilot, VS Code, PyCharm, Postman

EDUCATION & CERTIFICATION

- BS Computer Engineering, University of the Immaculate Conception (2014)
- AWS Certified Cloud Practitioner
- AWS Certified Machine Learning Engineer – Associate (in progress)